

Reinforcement learning for optimizing real-time interventions and personalized feedback using wearable sensors

Jyotsnarani Tripathy^{a,*}, M. Balasubramani^b, V. Aravinda Rajan^c, Vimalathithan S^d, Anurag Aeron^e, Meena Arora^f

^a Department of CSE-AIML & IoT, VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad, Telangana, India

^b Department of Computer Science and Engineering, VSB Engineering College, Karur, Tamil Naud, India

^c Department of Computer Science and Engineering in Kalasalingam Academy of Research and Education, Krishnankovil, Srivilliputtur, Tamil Nadu, India

^d Department of CSE, Mohamed Sathak AJ College of Engineering Chennai, Tamil Nadu, India

^e Department of Computer Science and Engineering, Meerut Institute of Engineering & Technology, Meerut, India

^f Department of IT, JSS Academy of Technical Education, Noida, U.P, India

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ABSTRACT

Many countries have seen a steady increase in lifetime duration through the past many years as a result of notable advancements in ecological and individual cleanliness, general health, including healthcare. Hence, it is anticipated that rising life duration and declining birth levels will result in a sizable aging population in the foreseeable future, placing a heavy strain on such nations' societal structures. For the benefit of senior medical and wellness, it is crucial to create affordable, user-friendly technologies. Older people can remain in their cozy homes rather than costly medical centers due to mobile health surveillance, which depends upon portable, non-intrusive devices, and actuators, as well as contemporary interaction and data innovations. It is an operational and economical approach. This work investigates how electronic devices can be used to optimize in-the-moment actions and deliver customized feedback using reinforced learning. By utilizing the information obtained from such sensors, the suggested structure dynamically modifies solutions according to each individual's reaction, increasing the efficacy of customized input. The reinforced method of learning optimizes the course of action for a variety of circumstances by interactively improving its tactics. By introducing an innovative strategy to improve immediate responses and provide customized input, this study advances the nascent field of wireless technologies as well as artificial learning and eventually raises the effectiveness of customized healthcare and well-being apps.

1. Introduction

About forty percent of avoidable fatalities are caused by long-term illnesses and inadequate health behaviors, which also have negative impacts on financial markets and standards of existence. These days, chronic illnesses are such common that they account for over 40 million mortalities annually, or 70 percent of every fatality worldwide [1]. The ability of individuals to handle their wellness is crucial when it comes to treating chronic illnesses and unhealthy patterns and behaviors. Following food regimens, taking prescription drugs as directed, or engaging in physical activity are a few examples of similar self-managing practices [2]. To effectively assist persons in these

operations, it is important to take into account their specific objectives and habits, as well as their emotional and emotional settings and contextual circumstances [3]. Numerous researches have demonstrated that individualized self-management guidance for prevalent medical conditions or chronic illnesses is more effective than person-invariant methods [4,5].

Digital approaches are an excellent way to help individuals with their self-monitoring tasks. Current technical developments in the portable, implantable, and consumable medical sensor domains [6] have made it possible to temporarily give adaptable therapies via handheld gadgets as needed. In such a way, the primary goal of this study is to provide a method that trains how to give personalized

* Corresponding author.

E-mail addresses: jtjyotsna@gmail.com (J. Tripathy), balasubramanimvbs@gmail.com (M. Balasubramani), v.aravindarajan@klu.ac.in (V.A. Rajan), snvimalathithan@gmail.com (V. S), anuragaeron.iitr@gmail.com (A. Aeron), meenaarora@jssaten.ac.in (M. Arora).

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interventions while taking into account the short- and long-term circumstances of individuals. Concerning actual healthcare programs, the goal is to enhance commitment by avoiding the needless strain of such treatments, ultimately leading to improved medical results.

Various researches have been conducted regarding the use of biological signals to identify and forecast psychological states during regular ordinary circumstances as well as regulated research environments using outside influences [7–9]. Meanwhile, as this study will show afterward, the last step is an extremely difficult undertaking, mostly because of the obstacles in gathering information as well as data labeling. However, because people vary in their biology, habits, and the variety of tasks they participate in, bio-signaling has different patterns throughout normal contexts. Furthermore, there is discrimination in the descriptions gathered since various people have various understandings of strain. As we demonstrate there will be a great need to personalize prediction algorithms to users based on the wide diversity of behaviors they participate in. Meanwhile, there are cognitive as well as technological difficulties because these decisions must be made at the implementation stage.

Since environmental momentary analysis (EMA), which is disclosed strain, is regarded as the benchmark for quality, the consumer provides the basis for the data required for personalization. As a result, the consumer must reply to inquiries at the moment by connecting the information gathered through the detectors to emphasize markers in routine situations. To fulfill such requirements, the framework must be able to gather environmental factors and biological signals, generate with gather EMAs, as well as rebuild as well as update the framework. Consumer commitment remains a challenging element with these kinds of structures, though. The observed individual might not reply to numerous inquiries for labeling, which will raise the possibility of abandonment. Furthermore, this is possible that people will respond to EMAs slowly or not at all as they go about their regular lives, which will decrease the number of tags available in addition to the relationship between them as well as the biological signals.

2. Literature review

We examine similar projects from two major angles. We initially examine research that offers individuals with chronic illnesses or other health issues individualized help (via mobile phones initiatives). Their customization strategies and systemic architecture fascinate us. Secondly, we describe research that suggests several approaches to implementing transferable knowledge across instructional settings and reinforcing prior activities.

Personalization of activities through flexibility and prompt delivery: Numerous research studies have examined providing individuals with medical conditions with customized support [10–12]. A system featuring many self-managing supporting elements was created by Gustafson et al. to help people overcome alcohol dependence [13]. Each of their method's elements, the GPS tracking element, alerts users if they go near a predetermined location. The approach put forth by Waki et al. analyses twice-daily patient data on diet and activity and then produces a measure in response [14]. Different fields, like hospital learning, or personalized healthcare [15,16], use reinforcement learning for personalization. Regarding how these mechanisms decide which treatments to give, we regard them as stable. Unlike other methods, ours maintains an array of adaptive decisions within the constraints that comprise the RL process. This offers a methodical approach to enhance or modify the provision of interventions dynamically over time in response to shifting individual situations.

Numerous studies in research have looked into the optimization of providing provisions concerning just-in-timeliness as well as flexibility. A few of them focus on certain illnesses. For instance, the determinants of selecting food with overweight are found using the agent-based approach created by Chih et al. [17]. According to an identical manner, Goldstein et al.'s approach anticipates nutritional failures and

provides individualized remedies [18]. Their approach finds significant signs of possible breakdowns by using controlled machine learning approaches on physical contextual factors. However, research is also being done on more general solutions that constantly optimize JITAI elements. By detecting long-term alterations in the framework's variables and modifying the intervening factors appropriately, these frameworks put into practice computational representations of current behavior transformation concepts [19]. For example, the Concept of Expected Behaviour [21] is realized by the mechanism that Navarro-Barrientos et al. presented [20]. The technique's goal is to modify interference variables in a way that helps customers reduce obesity. The Social Cognition Theory [22] helps individuals perform their exercising operations, which is beneficial for Martin et al.'s approach [23]. A general theoretical framework for creating data-oriented computational frameworks with JITAI is presented by Spruit et al. [24].

The word "stress" has access to both interior (such as personality types, approach to contemplating, challenges in relationships, finances, etc.) including exterior (such as style of life, interpersonal issues, etc.) issues that cause undesirable feelings (such as anxiety, dread, etc.) and related physical alterations. The term "physical stress responses" describes the physical alterations brought on by external stressors, which are actions or circumstances. Stress's intricate structure may be divided into simple three primary parts: emotional, behavior-related, and biological. Historically, expressed questionnaires have been used to quantify anxiety. The Psychological Stressed Scaling (PSS), for instance, was created by S. Cohen et al. to gauge strain levels during the previous month. According to expressed inquiries, Holms as well as Rahe established an index that calculates the overall level of strain experienced over the previous year.

3. System model

3.1. Data gathering

Rather than using equipment designed specifically for analysis, we utilized an advertising wristwatch, Samsung's Galaxy Gearing Sports, for gathering information for this investigation. To obtain unprocessed information through the watch, we created an assortment of features and uses (1) a smartwatch application that gathers fragments of information using detectors. (2) a data synchronization application. (3) A scheduled application that uses the two technologies mentioned previously at predetermined intervals. (4) A traditionally scheduled services restart the programme. (5) A program that manually synchronizes every last bit of content.

3.2. Signaling

The subsequent detectors provide us with some important information:

PPG, or photoplethysmography: The measurement of lightning refraction through the skin's outer layer is provided by this detector. This pulse has been captured at 20 Hz, which is the highest rate a wearable device can handle.

Accelerometer (ACC): This device measures velocity at every angle and provides three readings. Twenty Hz is commonly used to analyze this indication.

Gyroscope (GYR): Gyroscopes use three directions for determining the angle of rotation. We also fixed the sensor's capturing frequency to around 20 Hz.

Pressure: The environmental pressure is measured at 10Hz by a humidity detector.

Our information-gathering function is configured to run continuously for 2 min, while the scheduling function is programmed for calling this function every 15 min. When combined with regular typical usage, the circumstances extend the battery's life to approximately two whole days. We see that with typical smartwatch use, batteries endure for

roughly 3.5 days. Every 6 h, our synchronization services were contacted to address unsuccessful synchronizations. Every sensing measurement and its timing are combined with an indication database using information gathered throughout the collecting interval.

3.3. Coordination

The collecting application will try to send the indicators' database straight to our website host via internet access upon saving it. The communication will try to transmit information via the cellphone utilizing the interface gadget when the wristwatch does not have internet access. That's going to use the Wireless lower-energy connectivity among the wristwatch and cellphone to build an imitation for the cellphone's Online application. The information is stored in temporary documents within the wristwatch by the collecting use, which ends if either of the two routes fails. All previously unsuccessful synchronization file deliveries will be retried upon activation of the synchronization program. The overview of the proposed methodology is shown in the block diagram below (Fig. 1).

We give a quick synopsis of Q-Learning as well as Deeper Q-Learning then direct a curious viewer to some of the several sites in the scientific field which go into great detail about this collection of techniques. The method's goal is to choose an activity a depending upon the method's present condition s at that moment. A statistic that gauges a state-action blending's efficiency (Q-value) serves as the technique's main operator:

$$Q : S \times A \rightarrow \mathbb{R}. \quad (1)$$

Q-values show the projected future advantage (or expense) of specific course steps in the context of the existing situation. In classic Q learning, while an individual communicates via the framework, a database comprising the Q-values mapping onto the (s, a) pairing is initialized and modified according to a predetermined pattern. The Bellman calculation, that refreshes the Q functions repeatedly as below:

$$Q^{new}(s_t, a_t) = (1 - \alpha)Q(s_t, a_t) + \gamma \left(R_t + \max_a Q(s_{t+1}, a) \right) \quad (2)$$

$$S^{IS} = \text{habit_strengths, behavioral_frequency, day_type, remembrance_behavior} >$$

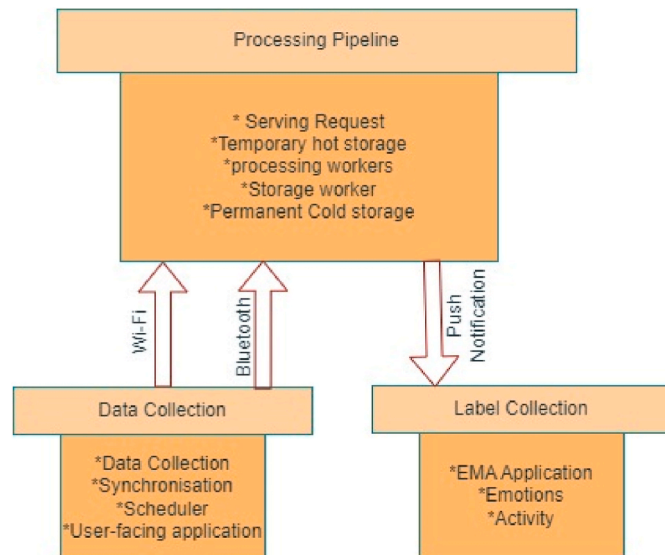


Fig. 1. Overview of the proposed model.

whereas t represents a time-based index, R_t seems the incentive at moment t , and α seems the developing pace which regulates the convergent velocity by expanding the effect of newly discovered data compared to earlier discovered data on the last Q function, has the most widely used version of this principle. It looks like that: A little γ encourages a myopic person to just think about present or short-term gains, whereas a greater γ encourages the participant to aim for permanent earnings. $\gamma \in [0, 1]$ represents the discounting rate that affects the significance of prospective profits. The procedure having the greatest Q-value in every condition s_t will be selected by the strategy (during the after-training exploiting stage):

$$a_t = \operatorname{argmax}_a Q(s_t, a) \quad (3)$$

In Deeper Q-Learning, neural networks are used for estimating the Q-values rather than a list of Q ratings. A popular technique seems for implementing the Q system as an entirely linked neural system using N (various state attributes) inputs as well as M (the total number of prospective behaviors) result nodes. Every result point gives the predicted Q-value associated with its related activities. This method works well when the behaviors are restricted to just a few distinct numbers, such as binary behavior. Next, the neural networking gets trained as well and the system's parameters are optimized using the Bellman formula (Equation. 2). The condition, activity time, incentive operation, and networking structure that we employed in our application are all defined in the sections afterward.

3.4. Intervention-selecting method

The intervention-selecting algorithm's MDP will be demonstrated as follows:

$$M^{IS} = \langle S^{IS}, A^{IS}, P^{IS}, R^{IS} \rangle.$$

The state-based group is called S^{IS} , and every condition, $s^{IS} \in S^{IS}$, appears by a pair made by combining the following four factors:

For this approach, as explained next, we refrained from using state variables having constants, just like in the opportune-momentary-detection scenario: • *habit_strengths* = $\{x : x \geq 0 \text{ and } x < 10, x \in \mathbb{N}\}$. The capability of an automated action without requiring mental effort for recalling to carry out the behavior is referred to as habitual intensity. Put differently, the stronger the actions, the fewer signals or actions are needed to prompt the person to act. • *behavioral_frequency* = $\{x : x \geq 0 \text{ and } x < 10, x \in \mathbb{N}\}$. The proportion of acting recurrence to all possibilities to conduct an action, during a particular period in the previous history, gets how many instances an action is executed. • *day_type*: This enumerate accepts entries for either weekdays or a holiday. The significance is derived using the replicated period, which corresponds to the time for the training session. • *remembrance_behavior*: Another list that accepts positive or negative entries for retaining behavior. If the subject could recall carrying out the behavior according to the strategy, it will be indicated by the particular quantity within this variable.

The following represents the definition of the policy-selecting algorithm's activity setup:

$$A^{IS} = \{J_1, J_2, \dots, J_n, No_Intervention\}$$

One of the policy kinds within the collection seems $J_1, J_2, \dots, J_n$, which

is meant to be recognized by subject matter specialists when tackling a specific illness or health issue [35]. Apart from the designated involvement categories, the option to provide no support at all, denoted by the $No_Intervention$ tag, remains conceivable. An appropriate number of actions (EA_a) with every task ($a \in AP$) specified in a strategic program is associated by an inner converting (M) in the following way:

$$M : A \rightarrow EA_a, a \in AP, EA_a \in CA^S$$

EA_a is subsequently reduced by taking into account the action kinds that qualify for selecting sites. Only its warning actions are taken into consideration at decision-making stages that come before the scheduled action. In any other case, only motivational therapies qualify. In this regard, we designate the last collection of eligible activities as EA_{DP} . A strategy is_{a_t} is chosen via such collection, or $is_{a_t} \in EA_{DP}$, at every choice step within the intervention-selection framework.

Three factors influence R^S feature: recalling the behavior, whether or not a measure is given, then whether or never a solution is engaged upon. Such factors are used to determine the incentive at every point of transition (is_{s_t}), phase t . The incentive function, as shown in Equation. (3), is composed of a variety of constraints on such factors.

Using the calculation, the parameters for all six corresponding circumstances will appear as $-50, -10, -5, -3, -1$ as well as 10 . The idea behind this kind of implementation has to provide comparatively greater incentives in situations where the individual chooses a response kind as a countermeasure ($remembrance_behavior = \text{untrue}$ as well as $is_{a_t} \neq No_Intervention$) as well as will not recall carrying out the behavior. Positive reinforcement is given for participating in the action ($behaved = \text{true}$), which makes the actions that an individual chooses not to participate in less likely. The technique's worst choice is going to choose no approach at all, particularly in cases when the subject will not recall engaging in the action ($remembrance_behavior = \text{untrue}$ as well as $is_{a_t} \neq No_Intervention$). It refers to the situation wherein the biggest reward in elevation, however, an unfavorable one—is produced.

P^S constitutes a vector of possibilities, $P(is_{s_{t+1}} | is_{s_t}, is_{a_t})$, that represents the likelihood that an activity is_{a_t} will be taken to change the condition via is_{s_t} into $is_{s_{t+1}}$. Using a sequential learning method, the intervention-selecting system assigns a daily value for every event. The strategy for action linked to the individual lays out condition changes during a period or every day. The routine development framework integrated with the intervention-selecting framework computes successive state variables. As a result, predictable transitional behaviors exist between states is_{s_t} and $is_{s_{t+1}}$.

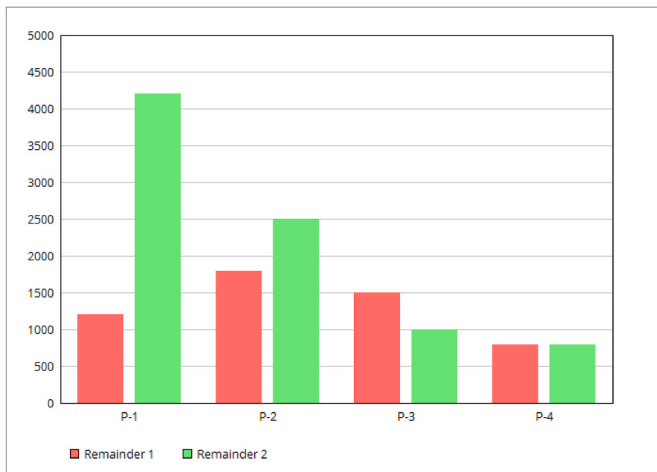


Fig. 2. Person Vs number of delivered Interventions.

4. Result and discussion

Two tests are carried out, the current State Classification is not used in the initial test since it hasn't been trained properly. The State Classification is trained using the information produced from the initial test. The State Classifiers are used throughout the subsequent study to choose responses. The subsequent study generated outcomes that are shown throughout the following subsection.

A lower commitment level translates into lesser habit power, based on the pattern-forming paradigm. Due to such, this is possible to see that throughout the initial stage, Individual-1 typically received more interactions overall than the remaining participants. In other words, the greatest amount of initiative is given to Individual 1, who displays the least attachment level. Fig. 2 illustrates that Individual 1 received greater attention overall than the other participants. This refers to normally predicted, as Individual-1 should receive additional reminders to carry out the behavior along the entire procedure. However, after some time, the various operations supplied become nil. The variety of provided actions varies in response to the intensity of each routine; while the routine grows stronger, fewer measures are supplied, and this pattern continues during the experiment.

Quantity of remedies administered for each kind: Based upon their classifications, all three modeled interaction kinds are grouped into two distinct groupings. That is, operations of one group will be substituted only by only operations from an identical group. Therefore, every group's outcomes will be separately assessed.

Remembrance -1 , Remembrance -2 , and Encouragement are recall actions. Their function is to remind. Likewise, the motivational group's function is to encourage. As previously said, every unique desire rating is utilized as an estimate of interaction while the consumer is shown the appropriate form of interaction. Individual 2 has an 80 percent probability of reacting to Remembrance -2 or Inspiration when provided via it, whereas Individual 1 does not respond to Remembrance -1 . We cannot anticipate a normal association among persons for a specific solution class as well as delivery rates due to the discontinuous selection chances. Even when the associated preferred chances remain identical, these proportions for an identical individual could not be connected. Remembrance -2 and Inspiration, for instance, have identical proportions for Individual-4; however, because Remembrance -1 may be given as an alert action, this can't be anticipated that those two initiatives will achieve a similar proportion.

In Fig. 3, we analyze the impact rates within each unique group upon a personal base. The original preferences per individual are reflected in the proportions of Remembrance-1 as well as Remembrance-2. For instance, given that Individual-3's choices are 10% during Remembrance -2 as well as 80 percent for Remembrance -1 , the proportion of

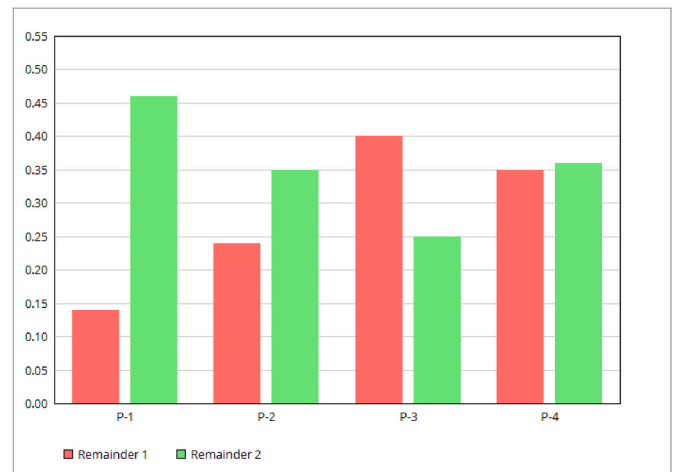


Fig. 3. Person vs ratio of the delivered interventions.

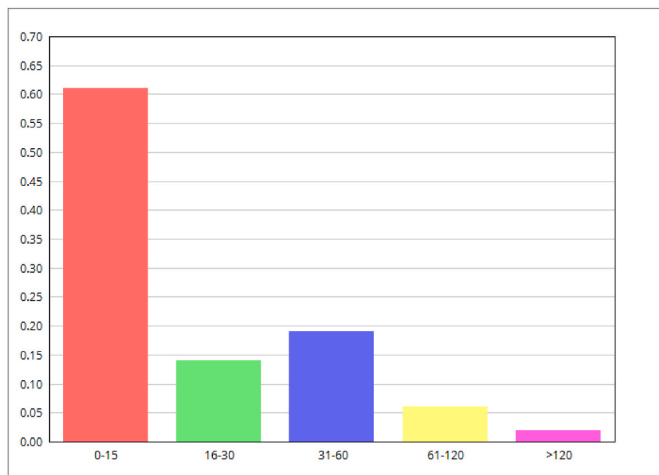


Fig. 4. Difference in time between the intervention delivery and behavior performance.

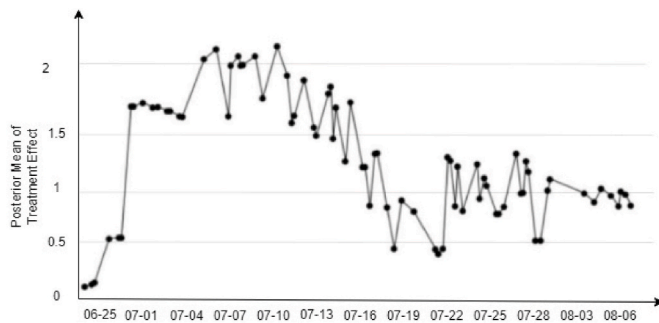


Fig. 5. Time vs Posterior mean of treatment effect.

Remembrance -1 with Individual-3 is greater compared to the proportion of Remembrance -2. Remembrance -1 and Remembrance -2 have proportions that are nearly identical to one another.

However, it remains possible to find discrepancies between desires and the outcomes of the initial view. Individual-1, for instance, is not at all interested in Remembrance-1 and is not interested in Inspiration. However, the data indicates that the rates for Inspiration as well as Remembrance-1 remain at 40% as well as 15%, respectively. The primary cause of the discrepancy in Remembrance-1 involves the development agent's arbitrary choice of interactions in unknown states, especially ones it has never experienced before. In other words, the training agents choose between Remembrances 1 and 2 with arbitrary. Such an issue also pertains to commitment. Furthermore, if it is anticipated that the individual would not engage in the behavior, the method prioritizes the execution of any kind of response. Any kind of interaction will serve as an indication in these situations.

Spatial convergence of behavioral execution and solution provision: The simulation's structure suggests that behavioral execution and solution provision will not always occur simultaneously. Almost all of the moment, the two are separated in space. To maximize the impact of these reminders, the intended action seems to provide it as shortly as feasible to an occasion where the circumstances are right to carry out the behavior. The proportion of actions falling within the pre-established limits of timing variations is displayed in Fig. 4. In a graph, a bar represents every category. As a period disparity grows, the proportion of provided interventions generally drops. Just at the following bar, or that 31–60-min bar, does the sequence break. The everyday tasks created during the point frame are inappropriate for interacting via intervention, which constitutes the cause. The fact that the approach lacks suggestions in stages when they are inappropriate for interaction seems

a sign of its capacity to develop customized behaviors.

Proceeding, we will look at participant ID = 4, who seems to be responding quite well to the suggested activities. Initially, we observe that throughout this instance, the possibility of slicing occurs more than once. That exists, a mean limitation upon various proposals every day-time is reached since the randomized possibility surpasses the limitation of $1 - \epsilon_0 = 0.8$. Subsequently, the diagram's left-hand bar indicates that our RL method's randomized possibilities don't truly begin to decline till 07–16. The proximal significance is expected to react with the increased dosage immediately and indicate that there will be a greater reduction in probabilities.

Look at the left-hand chart in Fig. 5. It shows a graph plotted between time versus posterior mean of treatment effect. Comparing the black highlights, which represent the real randomizing possibility, against the red highlights, which represent the randomizing possibility despite the proximal values adaptation, to observe how the proxies are, via reality, decreasing the possibility of providing the pedestrian proposal while a postponed impact is provided. Under the optimal scenario, there should be a greater difference around the black as well as red dots around 07–16 as well as 07–15.

5. Conclusion

It is difficult to identify physical strain along with various markers of mental health solely on physical information collected from natural environments, because of distortion in both signaling and tags. Forecasting accuracy is greatly improved when every individual's recognition systems are personalized. We created a multiple-layered infrastructure to manage online database gathering that transmits as well as analyzes biological signals instantaneously. To enhance user interaction as well as enable customization, we have created a self-learning agent depending upon Q-learning that can recognize influential instances and record individuals' responses according to the situation. We intend to make public the collection of medical information, health, and behavioral tags that we gathered over an extended period in natural environments. Our suggested context-aware engaged learning agents minimize the various inquiries needed to achieve the intended degree of customization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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